


 Course & Exam
Particulars

Course Type: Core

Date:

Course Title: Artificial Intelligence

Time:

Course Code: 22AD42

Section: A

Maximum Marks: 50

Duration: 01hr 30mins

UG Degree | Mid-Semester Examination-1 | AY: 2025-26 (EVEN)
Department of Artificial Intelligence and Data Science

Question No	Question	Marks	CO; BL
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Instructions:

1. Answer the following questions.
2. Programmable calculators and electronic gadgets are not permitted.
3. Duration and maximum marks are as indicated on the question paper.

UNIT-I

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|---|---|--|---|---------|
| 1 | a | Discuss in detail the History of AI | 6 | CO1, L2 |
| | b | Explain the Depth first iterative deepening | 7 | CO1, L3 |
| | c | Interview the steps and features of State Space Search | 7 | CO1, L3 |

OR

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|---|---|---|---|---------|
| 2 | a | Differentiate between BFS and DFS | 6 | CO1, L2 |
| | b | Illustrate the Heuristic Search Problem with an example | 7 | CO1, L3 |
| | c | Identify in detail Hill climbing with an example | 7 | CO1, L3 |

UNIT-II

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|---|---|---|---|---------|
| 3 | a | Discuss about the Random Walk Algorithm | 6 | CO2, L2 |
| | b | Infer about the Simulated annealing with an algorithm | 7 | CO2, L3 |
| | c | Build the Eight Queen's problem | 7 | CO, L3 |

OR

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|---|---|---|---|---------|
| 4 | a | Illustrate the concept of Ant colony optimization | 6 | CO2, L2 |
| | b | Outline in detail Branch and Bound | 7 | CO2, L3 |
| | c | Inspect the A* algorithm with an example | 7 | CO2, L4 |

UNIT-III

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|---|---|--|---|---------|
| 5 | a | Discuss the Results related to Admissibility of A* | 5 | CO3, L2 |
| | b | Construct the Iterative Deepening A* | 5 | CO3, L3 |

OR

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|---|---|--|---|---------|
| 6 | a | Interpret the advantages and Disadvantages of IDA | 5 | CO3, L2 |
| | b | Examine the Board Games and Game Trees in Game playing | 5 | CO3, L4 |

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